Points 100 Not Published

Details Questions Mastery Paths

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Conceptual

Group Name Pick 10 questions, 3 pts per question Pick questions, pts per question

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Questions will be pulled from the bank: **Exam 2 - Conceptual**

Question 3 pts

VeloShip Fulfillment's operations manager tracked customer complaints over the last quarter:

Cause	Frequency
Wrong item shipped	48
Late delivery	35
Damaged packaging	22
Missing item in order	11
Incorrect address label	4
Total	**120**

Using Pareto analysis, select all causes VeloShip should prioritize to address at least 80% of complaints.

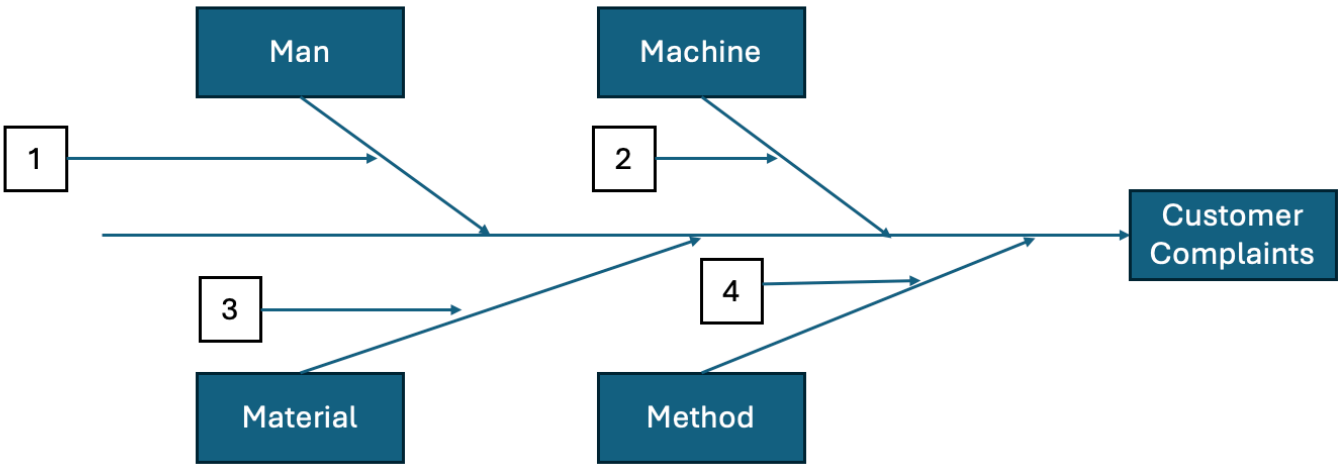
- [ ] Wrong item shipped
- [ ] Missing item in order
- [ ] Incorrect address label
- [ ] Damaged packaging
- [ ] Late delivery

Question 3 pts

The manager of Smith, Schroeder, and Torn is investigating root causes of customer complaints following a series of problem deliveries. Issues identified include: truck broke down, ran out of packing boxes, multiple deliveries in one day caused the truck to be late, no furniture pads, employee dropped several items, driver got lost en route, ramp into truck

was bent, no packing tape, new employee doesn't know how to pack, moving dolly has a broken wheel, and employee was late to work.

The cause-and-effect diagram below organizes these issues into four categories — **Man, Machine, Material, and Methods**— as shown in the figure provided.



Match each cause below to the correct category bone.

New employee doesn't know how to pack

1
Moving dolly has a broken wheel

2
Multiple deliveries scheduled on one trip

4
No packing tape available

3

Question 4 pts

NeoZapato operates two production facilities. Last week's data is summarized below. Each pair of sandals sells for $40.

	Tijuana Facility	Los Angeles Facility
Pairs produced	1,600	2,100
Labor hours worked	320	350
Labor cost ($)	$3,840	$15,750
Materials cost ($)	$2,800	$5,200

Overhead cost ($)	$1,200	$3,000

1. Which facility has higher labor productivity? [lp]

2. Which facility has higher multifactor productivity? [mp]

Show Answers for lp

LA Facility
Tijuana Facility

Question 5 pts

TechAssemble's quality inspector selects 4 motherboards per shift and measures a critical dimension (mm). Data from 5 shifts is shown below. Constants for n = 4: A_2 = 0.729, D_3 = 0, D_4 = 2.282.

Shift	x_1	x_2	x_3	x_4
1	10.4	10.2	10.5	10.3
2	10.1	9.8	10.3	10.2
3	10.6	10.7	10.4	10.5
4	10.2	10.3	10.1	10.4
5	10.5	10.3	10.4	10.6

Calculate the $\overline{\overline{X}}$-chart control limits. UCL = [ucl], LCL =[lcl]

Show Answers for ucl

10.588
10.593
10.528
10.632

Question 4 pts

VeloShip Fulfillment runs a Kanban loop between its Pick & Pack station and Quality Check station. The Quality Check team consumes 480 order containers per day. Once an empty container is sent back to Pick & Pack, it sits on the transfer cart and waits in the queue for 2 hours before anyone begins refilling it. Once work begins, it takes 30 minutes to process and fill one container. Each container holds 24 orders. Management applies a 10% safety factor to account for demand variability. The facility runs one 8-hour shift per day.

How many Kanban containers should circulate in this loop?

5
6
7
8

Question 4 pts

TechAssemble's quality team inspects a random sample of 150 motherboards per shift and records the number of defective units. Results from the last 10 shifts are shown below.

Shift	1	2	3	4	5	6	7	8	9	10
Defectives	5	3	8	2	6	16	4	2	7	7

Select the appropriate chart type, calculate the control limits, and determine whether any shift is out of control.

- UCL = 0.088, LCL = 0; Shift 6 is out of control
- UCL = 0.088, LCL = 0; all shifts in control
- UCL = 0.096, LCL = 0; all shifts in control
- UCL = 0.092, LCL = 0; Shift 6 is out of control

Question 4 pts

Riverside Medical Center processes patient discharges through four sequential steps. The discharge unit operates one 10-hour shift per day and handles 40 discharges per day.

Step	Cycle Time	WIP (patients waiting)
Medical Assessment	8 min	8 patients
Discharge Planning	10 min	12 patients
Pharmacy & Scripts	6 min	20 patients
Final Clearance	5 min	16 patients

Calculate the takt time, total processing time, and total production lead time.

- Takt = 15 min; Total PT = 29 min; Total Lead Time = 1.40 days
- Takt = 12 min; Total PT = 29 min; Total Lead Time = 1.40 days
- Takt = 15 min; Total PT = 29 min; Total Lead Time = 1.00 days
- Takt = 12 min; Total PT = 29 min; Total Lead Time = 1.75 days

Question 5 pts

PeakForm Athletic manufactures two lines of yoga mats on identical molding machines. The plant runs one 8-hour shift per day, 250 days per year, and maintains a 25% capacity cushion.

	Standard Mat	Pro Mat
Annual Demand	15,000 units	7,000 units
Processing Time	8 min/unit	18 min/unit

Average Lot Size	50 units/lot	35 units/lot
Setup Time per Lot	30 min/lot	60 min/lot

How many machines does PeakForm require?

2 machines
3 machines
4 machines
5 machines

Question 7 pts

Ridgeline Furniture allocates production time on its constrained CNC router (2,400 min/week) across three product lines.

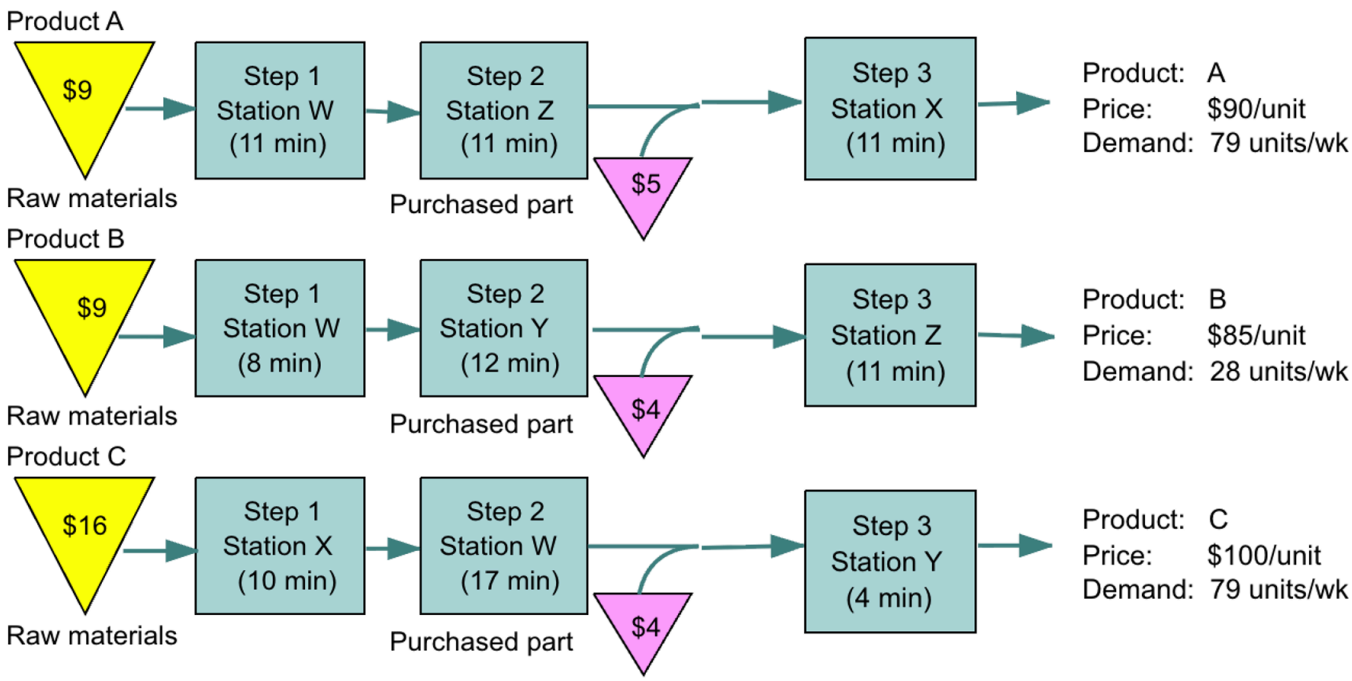
Product	Contribution Margin	Bottleneck Time (min/unit)	CM per Bottleneck Min	Weekly Demand
Dining Table	$76	11	$6.91	79 units
Bookshelf	$72	8	$9.00	80 units
Cabinet	$80	17	$4.71	60 units

Using the bottleneck method, how many units of each product should Ridgeline produce weekly?

Bookshelf = 80, Dining Table = 79, Cabinet = 52
Cabinet = 60, Dining Table = 79, Bookshelf = 63
Bookshelf = 80, Dining Table = 68, Cabinet = 60
Cabinet = 60, Bookshelf = 80, Dining Table = 71

Question 7 pts

York-Perry Industries (YPI) manufactures three guitar models (A, B, C) across four workstations (W, X, Y, Z), each with 2,400 minutes available per week.



Using the product routings and demand data shown in the figure, which workstation is the bottleneck and what is the load?

Workstation X, with a total load of 1,869 minutes
Workstation W, with a total load of 2,436 minutes
Workstation W, with a total load of 2,520 minutes
Workstation Y, with a total load of 2652 minutes

Question 5 pts

Clearwater Instruments assembles a flow meter with 7 work elements. The table below shows task times and precedence relationships. The line must produce 45 units per hour.

Work Element	Time (sec)	Immediate Predecessor(s)
A	40	—
B	30	A
C	50	A
D	25	B
E	20	B
F	30	C
G	15	D, E, F
Total	**210 sec**	

Calculate the cycle time, theoretical minimum stations, efficiency, and balance delay for this line.

c = 45 sec; TM = 5 stations; Efficiency = 93.3%; Balance Delay = 6.7%
c = 80 sec; TM = 3 stations; Efficiency = 87.5%; Balance Delay = 12.5%
c = 80 sec; TM = 4 stations; Efficiency = 65.6%; Balance Delay = 34.4%

c = 80 sec; TM = 3 stations; Efficiency = 75.0%; Balance Delay = 25.0%

Question 7 pts

ClearFlow Manufacturing assembles industrial filtration units on an assembly line. Each unit requires 11 work elements (A through K), totaling 363 seconds of work. The table and precedence diagram below describe the task times and sequencing requirements.

Work Element	Time (sec)	Immediate Predecessor(s)
A	35	—
B	75	A
C	20	B
D	40	C
E	13	B
F	65	E
G	53	B
H	26	A
I	15	H
J	8	I, G
K	13	D, J, F
Total	363	

The line must produce 45 units per hour. Engineering has confirmed the cycle time is 80 seconds. Using the shortest work element rule, which tasks are assigned to each workstation?

S1={A,H,I} · S2={B} · S3={E,C,D} · S4={G,J} · S5={F,K}
S1={A,H,I} · S2={B} · S3={G,C} · S4={D,E,J} · S5={F,K}
S1={A,B} · S2={C,H,I} · S3={E,G} · S4={D,J,F} · S5={K}
S1={A,H,I} · S2={B} · S3={E,D} · S4={J,G} · S5={C,K,F}

Question 6 pts

Precision Parts Co. manufactures aerospace fastener brackets with tightly controlled mounting hole spacing. The design team has set a specification of 105.5 ± 0.3 mm for this dimension. During a recent process audit, the quality engineer measured the following:

- Process mean ($\bar{x}$): 105.6 mmProcess standard deviation (σ): 0.09 mm

Match each measure with its correct value.

Process Capability Ratio (Cp)

1.11

Process Capability Index (Cpk)

0.74

Maximum σ required to achieve 4-sigma capability (assuming current mean)

0.05

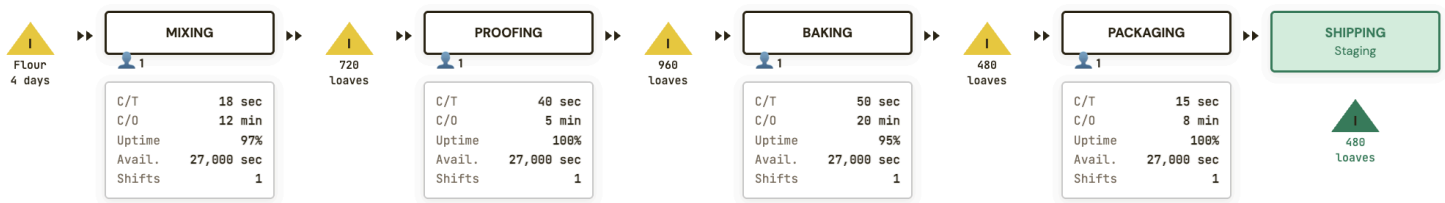
Other Incorrect Match Options:

- 1.33
- 0.89
- 0.08

Question 6 pts

SunCrest Bakery produces artisan sourdough loaves for wholesale grocery distribution. The operations manager, Maya Chen, has mapped the current state of the production line to identify waste and opportunities for improvement. The bakery runs one shift per day with 27,000 seconds of available production time. Customer demand is 480 loaves per day.

The production process flows through four sequential steps — Mixing, Proofing, Baking, and Packaging — before loaves are staged for shipping. Inventory levels between steps, cycle times, changeover times, and uptime percentages are shown in the value stream map below.



- ***What is the total processing time? [pt]***
- ***What is the total production lead time? [lt]***
- ***What is the process cycle efficiency? [pce]***

Show Answers for pce

0.048%

0.48%

0.058%

0.58%

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